

Partical 4 task 2

varshithcomple@VARSHITs-MacBook-Air - % nano ptask2.c

UN PICO 5.89 File: ptask2.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i, status;
    pid_t pid[3];

    printf("Parent Process ID: %d\n", getpid());

    // Create 3 child processes
    for (i = 0; i < 3; i++)
    {
        pid[i] = fork();

        if (pid[i] < 0)
        {
            perror("fork failed");
            exit(1);
        }

        if (pid[i] == 0)
        {
            printf("Child %d: PID = %d, Parent PID = %d\n",
                   i + 1, getpid(), getppid());

            sleep(3 - i);

            printf("Child %d completed.\n", i + 1);
            exit(i + 1);
        }
    }

    // Parent waits for specific children
    for (i = 0; i < 3; i++)
    {
        waitpid(pid[i], &status, 0);
        if (WIFEXITED(status))
        {
            printf("Parent: Child %d with PID %d finished with status %d\n",
                   i + 1, pid[i], WEXITSTATUS(status));
        }
    }

    printf("All child processes completed.\n");
    return 0;
}
```

Get Help
Exit

WriteOut
Justify

Read File
Where is

Prev Pg
Next Pg

Cut Text
UnCut Text

Our Pos
To Spell

```
varshithcomple@VARSHITThs-MacBook-Air ~ % nano ptask1.c
varshithcomple@VARSHITThs-MacBook-Air ~ % gcc ptask2.c -o ptask2
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
varshithcomple@VARSHITThs-MacBook-Air ~ % ./ptask2
Parent Process ID: 55244
Child 1: PID = 55245, Parent PID = 55244
Child 2: PID = 55246, Parent PID = 55244
Child 3: PID = 55247, Parent PID = 55244
Child 3 completed.
Child 2 completed.
Child 1 completed.
Parent: Child 1 with PID 55245 finished with status 1
Parent: Child 2 with PID 55246 finished with status 2
Parent: Child 3 with PID 55247 finished with status 3
All child processes completed.
varshithcomple@VARSHITThs-MacBook-Air ~ % █
```